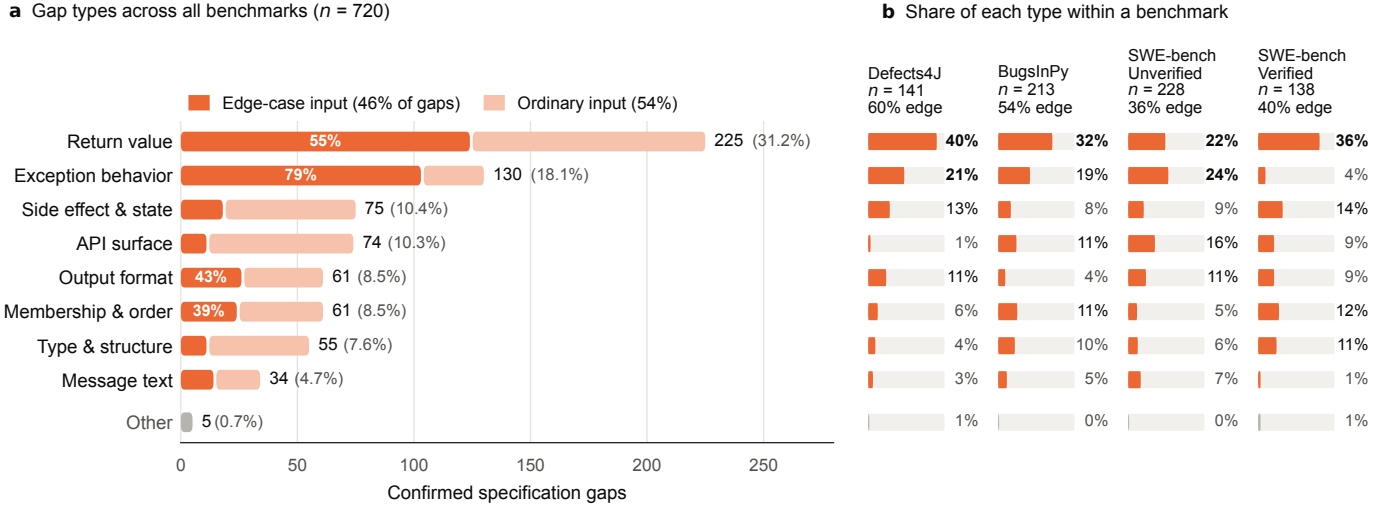
a Gap types across all benchmarks (n = 720)
Edge-case input (46% of gaps)
Ordinary input (54%)
Return value
55%
225 (31.2%)
Exception behavior
79%
130 (18.1%)
Side effect & state
75 (10.4%)
API surface
74 (10.3%)
Output format
43%
61 (8.5%)
Membership & order
39%
61 (8.5%)
Type & structure
55 (7.6%)
Message text
34 (4.7%)
Other
5 (0.7%)
0
50
100
150
200
250
Confirmed specification gaps
b Share of each type within a benchmark
Defects4J
n = 141
60% edge
BugsInPy
n = 213
54% edge
SWE-bench Unverified
n = 228
36% edge
SWE-bench Verified
n = 138
40% edge
40%
32%
22%
36%
21%
19%
24%
4%
13%
8%
9%
14%
1%
11%
16%
9%
11%
4%
11%
9%
6%
11%
5%
12%
4%
10%
6%
11%
3%
5%
7%
1%
1%
0%
0%
1%